Series circuits

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use current continuity and voltage drops in a single-path circuit.

SUCCESS CRITERIA - I can...

1. Identify a single current path.
2. Add series resistances.
3. Check that voltage drops add to the supply voltage.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Explain what a potential difference of **5 V** means in joules per coulomb.

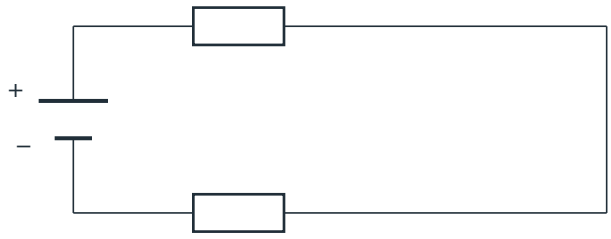
Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $R_{total} = R_1 + R_2$; $I_1 = I_2$; $V_{total} = V_1 + V_2$. Series resistors share one unbranched current path.

Worked reasoning: Two series resistors **2** and **4 ohms** on a **12 V** ideal supply have total **6 ohms** and current **2 A**. Drops are **4 V** and **8 V**; they add to **12 V**.

Your turn: Draw the loop and **label current** and both voltage drops.

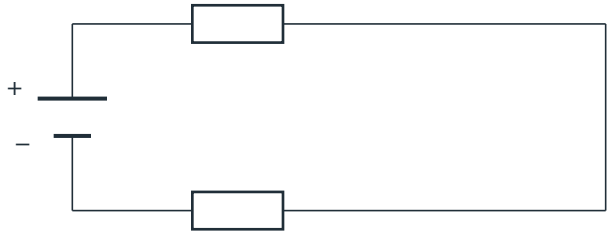


Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Trace the only current path in the series diagram. If either resistor connection opens, **what happens** to the current?

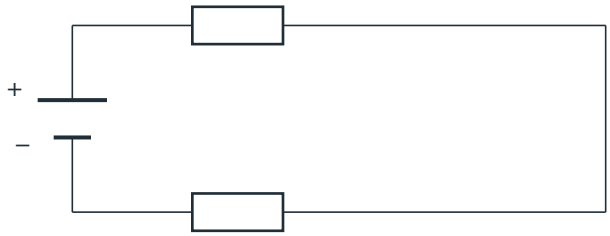


Circuit symbols: long/short lines = battery; rectangles = resistors.

Marzano check - circle: ① ② ③

04 Apply the physics

A **6 V** battery powers **2 Ω** and **4 Ω** resistors in series on a circuit board.
1. **Label both resistors** and the battery.
2. **Add resistances**, then **find the current through each** resistor.



Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

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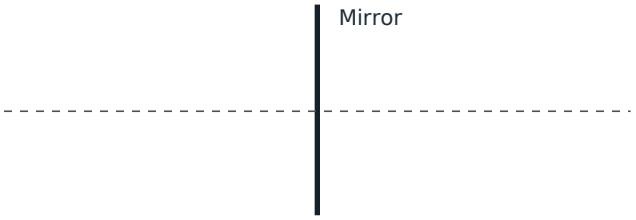
05 Explain your reasoning

A student assigns more current to the smaller series resistor. **Explain why this would imply** charge accumulating between resistors.

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A light ray strikes a plane mirror at **25°** from the normal.
Find the reflection angle and **label both angles** on a sketch.
Draw the rays and **label both angles** from the normal.

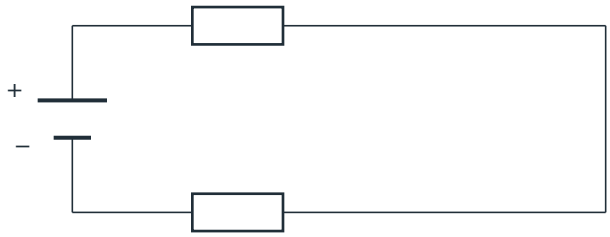


Add object / image or incident / reflected rays.

Marzano check - circle: ① ② ③

07 Exit check

In a **9 V** series loop, one component drops **3 V**.
1. **Find the other drop** if wires are ideal.
2. **Label the circuit** and **show the current path or** paths.

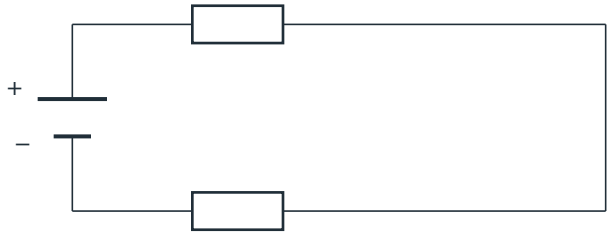


Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

Choose two positive series resistances totaling **10 ohms**. On a **10 V** supply, find current and each voltage drop. **Give a second valid design**. **Label the circuit** and **show the current path or** paths.



Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③